CTIS-U9

Reg. No. | | | | | | | | | | | [17CS4SP04]

FOURTH SEMESTER B.Tech EXAMINATION, JUNE – 2019

OPERATING SYSTEM BUILDING BLOCKS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions by choosing at least two questions from each part.
2. Missing data may be suitably assumed.
3. Use of machine design data hand book is permitted.

PART – A

1. a. Define operating system. Discuss the services provided by operating system. **8**

b. Discuss how Operating System operates. **6**

2. a. Explore the different states of processes and repository used to store process information with relevant diagrams. **8**

b. Explore the different types of multithreading models. **6**

3. a. Discuss how processes are created and terminated in UNIX Operating System. **7**

b. Consider 4 jobs with (arrival time, burst time) as (0, 5) (1, 2) (2, 8) (3, 4). Estimate the average turnaround and waiting time for the jobs using FCFS, SJF and RR (q=3) scheduling algorithms, which one is better and why? **7**

4. a. Explain implementation of the monitor mechanism using semaphores. **6**

b. Given a system with total resources of A(3), B(14) & C(12) with the following snapshot: **8**

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	0	1	0	0	1	1	5	2
P1	1	0	0	1	7	5			
P2	1	3	5	2	3	5			
P3	0	6	3	0	6	5			
P4	0	0	1	0	6	5			

i. Evaluate the need matrix

ii. Check the system for Deadlock

iii. If process p0 request for resources (1,1,1), is request can be granted? (Using Banker's algorithm)

PART-B

5. a. What is paging? Explain paging hardware with translation look-aside buffer. **5**

b. Given the five memory partitions 100KB, 500KB, 200KB, 300KB and 600KB. How each of the first fit, best fit and worst fit algorithms place process of 212KB,417KB,112KB and 426KB size. Which algorithm makes efficient use of memory? **5**

c. List the advantages and disadvantages of following disk scheduling algorithms **4**

i. SSTF ii. C-SCAN

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6. **a.** Demonstrate the working of page replacement algorithms using the following page reference string 1,2,3,5,2,3,5,7,2,1,2,3,8,6,4,3,2,2,3,6 and assume 3 memory frames. Calculate how many page faults would occur in the case of **6**

   i. FIFO

   ii. LRU

   iii. Optimal

   b. List and discuss system call (File access mode) for file system in UNIX. **8**

7. **a.** Explain different types of files and operations on files. **8**

   b. Discuss the goals and principles of protection in a modern computer system. **6**

8. **a.** Elaborate on various types of accounts on a UNIX system. **5**

   b. Describe what should be aware of while using root account. **5**

   c. Discuss commands for managing users and user groups. **4**